

 PERSONA Tim Transmission Planner Methodical Diligent Patient Conservative Inquisitive "Transmission planning is a science — generation planning is an art. Transmission planning has few degrees of freedom; physics dictates how power flows."				
EXPERIENCE 10 years	EDUCATION B.Sc. Electrical Engineering	STUDY HORIZON Long & Mid Term	PRODUCT KNOWLEDGE Medium	USER TYPE Experienced user who doesn't want to rely on deep product knowledge to do his job.
PROFILE Senior transmission planner at Big Utility, promoted on his command of the steady-state behavior of the power system. He operates on an annual cycle, updating system plans every year, and is an expert with his current tool suite. He knows the methodology could be improved with probabilistic and economic results — but isn't sure how to get there.			ANNUAL WORKFLOW 1 Update loadflow models for current & planned generation/demand 2 Run contingency analyses to find bottlenecks 3 Develop enhancements & run cost-benefit per option 4 Prepare reports & communicate plans to stakeholders	
GOALS - Create optimal long-term plans for reliability & deliverability - Obtain early buy-in on assumptions to minimize rework - Perform comprehensive analyses that leave nothing to doubt - Publish work to advance the industry		MOTIVATIONS - Discovering significant benefit through robust investigation - Seeing projects through from planning to budget approval - Influencing significant capital investments - Providing adequate system flexibility to operators - Avoiding overlooked issues that cause wide blackouts		CHALLENGES - Waiting on inputs from other departments - Changing regulatory environments - Choosing the right tools - Conservative management resistant to change
LIKES - Geographical representation of network topology - Capturing physical characteristics in the model - Confidence that batch runs complete successfully - Actionable feedback when batch runs fail - Minimizing manual data manipulation - Visualizing inputs and outputs		DISLIKES - Needing multiple tools for reliability & cost/benefit analysis - Hard-to-trace cause/effect between system elements - Interrupting work to evaluate new generator locations - Worrying about work over the weekend - Spending too much time customizing the model - Merging data from multiple sources & formats		RESPONSIBILITIES - Ensure future deliverability of demand requirements - Understand relationships between system elements - Build processes that cut error & time, increase insight - Mentor junior team members - Complete FERC filings
SKILLS & SPHERE OF INFLUENCE Power Systems Economic Analysis System-wide Impact Logical Problem Solving Rate Case Risk Management IRP Generation Interconnection Grid Resiliency		BRANDS & INFLUENCERS Industry conferences & publications; IEEE Tooling: Siemens PSS®E, PowerWorld Popular Mechanics, Car & Driver, Consumer Reports		TRENDS IMPACTING TIM - Decisions shifting from reliability-only to economic impact - Capital harder to enter rate base — must show end-user benefit